

Strong Convergence in the Finite Coxeter Numbers Game: Parabolic Length and the Complete Boundary Atlas

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Abstract

We fix a finite reduced crystallographic root system and fire a simple node only when its coroot coordinate is strictly positive. In the strictly dominant chamber, every legal play is a reduced word for the longest Weyl element and has $|\Phi^+|$ moves. For an arbitrary dominant wall position, with J its zero-coordinate set, we prove that every play ends at the same anti-dominant point, accumulates to $w_0 w_J$, and has the exact common length $|\Phi^+| - |\Phi_J^+|$. The element $w_0 w_J$ is simultaneously the unique shortest representative of the terminal coset $w_0 W_J$ and the maximum of the minimal right-coset representatives W^J . Zero, disconnected, and rank-one faces are included without perturbation.

1 Finite owner and firing convention

Let $\Phi \subset V$ be a finite reduced crystallographic root system, with simple roots $\{\alpha_i : i \in I\}$, coroots $\alpha_i^\vee$, positive roots Φ^+ , and Weyl group $W = \langle s_i : i \in I \rangle$. We use

$$A_{ij} = \langle \alpha_j, \alpha_i^\vee \rangle. \quad (1)$$

For a dominant weight λ , write

$$x_i = \langle \lambda, \alpha_i^\vee \rangle \geq 0. \quad (2)$$

Node i is legal precisely when $x_i > 0$. Firing it replaces λ by $s_i \lambda$ and therefore

$$x'_j = x_j - A_{ji} x_i. \quad (3)$$

A firing word $(i_1, \dots, i_m)$ accumulates on the left:

$$w_m = s_{i_m} \cdots s_{i_1}, \quad \lambda_m = w_m \lambda. \quad (4)$$

Equations (1)–(4) remove the sign, transpose, and word-order ambiguities that otherwise obscure comparisons between versions of the numbers game.

The game and its strong-convergence theory are classical. Mozes introduced reflection processes on graphs and Weyl groups [5]; Eriksson developed the Coxeter formulation and strong-convergence theorem [2, 3]. We claim no priority. Our purpose is to give a convention-locked finite-type proof, close every dominant boundary, and make the reconstruction independently executable. Standard finite Coxeter and root-system facts are sourced to Humphreys and Bourbaki [4, 1].

2 The reflection-sign mechanism

Let ℓ denote Coxeter length. The following observation is the local engine of the proof.

Lemma 1 (root-sign criterion). *Suppose the current position is $w\lambda$. Then node i is legal if and only if $w^{-1}\alpha_i$ is a positive root whose support is not contained in $J = \{j : \langle \lambda, \alpha_j^\vee \rangle = 0\}$. Every legal firing obeys*

$$\ell(s_i w) = \ell(w) + 1. \quad (5)$$

Proof. The current coordinate is

$$\langle w\lambda, \alpha_i^\vee \rangle = \langle \lambda, w^{-1}\alpha_i^\vee \rangle. \quad (6)$$

A positive coroot is a nonnegative integral combination of simple coroots. Its pairing with λ is zero exactly when the corresponding root lies in the subsystem $\Phi_J = \Phi \cap \text{span}\{\alpha_j : j \in J\}$; outside that subsystem it is positive. A negative root gives a nonpositive pairing. This proves the legality criterion. The standard root criterion for Coxeter length says $w^{-1}\alpha_i > 0$ exactly when (5) holds. $\square$

Theorem 1 (strict chamber). *If $x_i > 0$ for every i , every complete legal firing sequence terminates at $w_0\lambda$, accumulates to the longest element w_0 , and has exactly $|\Phi^+|$ moves.*

Proof. Here $J = \emptyset$. Lemma 1 makes every legal word reduced, so length strictly increases and termination follows from finiteness of W . At a terminal position every simple coordinate is nonpositive; hence the position is anti-dominant. A finite Weyl orbit meets the closed anti-dominant chamber in exactly one point, here $w_0\lambda$. Strict dominance makes the stabilizer trivial, so the accumulated element is w_0 . Finally $\ell(w_0) = |\Phi^+|$. $\square$

3 Walls as a parabolic quotient

Now retain all zero coordinates. Put

$$J = \{i : x_i = 0\}, \quad W_J = \langle s_j : j \in J \rangle, \quad \Phi_J^+ = \Phi^+ \cap \text{span}\{\alpha_j : j \in J\}. \quad (7)$$

The stabilizer of λ is exactly W_J . Let w_J be its longest element and define the minimal right-coset representatives

$$W^J = \{w \in W : w\alpha_j \in \Phi^+ \text{ for every } j \in J\}. \quad (8)$$

Lemma 2 (quotient lifting). *Starting at $e \in W^J$, legal firing edges are exactly the length-increasing left weak-order edges*

$$w \mapsto s_i w \quad \text{that remain in } W^J. \quad (9)$$

In particular every cumulative firing element is the unique minimum-length representative of its right coset modulo W_J .

Proof. The parabolic exchange lemma says that for $w \in W^J$ and $\ell(s_i w) = \ell(w) + 1$, either $s_i w \in W^J$, or the reflection conjugate to s_i by w lies in W_J . In root language the exceptional alternative is precisely $w^{-1}\alpha_i \in \Phi_J^+$. Lemma 1 excludes exactly that zero-coordinate alternative. It includes every other length-increasing edge, proving (9) inductively from the identity. $\square$

Theorem 2 (strong convergence and exact parabolic length). *For every dominant λ in every finite reduced crystallographic root system, every legal play terminates. Every complete play has the same:*

1. *terminal position, namely the unique anti-dominant orbit point $w_0\lambda$;*
2. *cumulative element $w_0 w_J$;*
3. *number of moves*

$$\ell(w_0 w_J) = |\Phi^+| - |\Phi_J^+|. \quad (10)$$

Moreover $w_0 w_J$ is the unique shortest representative of the terminal right coset $w_0 W_J$, and equivalently the unique maximum-length element of W^J .

Proof. By Lemmas 1 and 2, each firing strictly raises length in the finite set W^J , so every play terminates. A terminal weight has all simple coordinates nonpositive and is therefore anti-dominant. Every finite Weyl orbit has one point in the closed anti-dominant chamber, so all plays end at $w_0\lambda$.

Since W_J fixes λ , all elements taking λ to this terminal point form the right coset $w_0 W_J$. Lemma 2 says that the cumulative element is its unique minimum-length representative. The standard parabolic decomposition identifies that representative as $w_0 w_J$; it is also the unique maximum of W^J . Thus the cumulative element is choice-independent.

Every firing word is reduced, so its move count is $\ell(w_0 w_J)$. Inversion sets give $\ell(w_0) = |\Phi^+|$ and $\ell(w_J) = |\Phi_J^+|$. The length-additive parabolic factorization $w_0 = (w_0 w_J) w_J$ now proves (10). Finally $w_0 w_J \lambda = w_0 \lambda$ because w_J fixes λ . $\square$

4 Every degenerate face

The theorem specializes without limiting arguments:

face	exact conclusion
strict dominance	$J = \emptyset$, cumulative element w_0 , length $ \Phi^+ $
arbitrary wall	retain J exactly; the length loss is $ \Phi_J^+ $
zero vector	$J = I$, $w_0 w_J = e$, terminal zero, no legal moves
disconnected Φ	component games interleave; lengths add and terminals form the product point
rank-one positive	A_1 sends $x > 0$ to $-x$ in one firing
rank-one zero	A_1 at $x = 0$ has no legal firing
zero-coordinate rule	legality is $x_i > 0$, never $x_i \geq 0$
scope stop	affine and indefinite generalized Cartan systems are not covered

For a reducible system $\Phi = \bigsqcup_r \Phi^{(r)}$, one has $W = \prod_r W^{(r)}$, $\Phi^+ = \bigsqcup_r (\Phi^{(r)})^+$ and $W_J = \prod_r W_{J_r}^{(r)}$. Firings in distinct components commute; hence complete component plays interleave freely and (10) adds componentwise. This proves the disconnected assertion rather than treating one sample as proof.

The finite hypothesis is structural: the proof uses longest elements, finite length ascent, and a finite parabolic quotient. We make no affine, indefinite, Kac–Moody, noncrystallographic, or arbitrary generalized-Cartan extension.

Conclusion

Positive-coordinate firing on a finite crystallographic Weyl system is an exact parabolic quotient ascent. In the strict chamber this reaches w_0 . On every wall it instead reaches the unique minimal representative $w_0 w_J$ of the terminal coset, losing exactly the positive roots of the stabilizer subsystem and preserving choice-independent terminal state and length. Zero, reducible, and rank-one faces require no exception to the formula.

References

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